

Project Proposal: Scope of Work

EGH490-1 RESEARCH PROJECT 1

Assessment #	1
Unit Learning Outcomes Assessed	ULO1 Conduct a literature review to inform an engineering problem using survey, reference management, analysis, and synthesis at a mastered level. ULO2 Formulate a proposal for an engineering project that addresses client needs, management of time, resources, and communication at a mastered level.
Task Weight (%)	40%
Due Date	11:59pm Wednesday, Week 7
Nature of Submission	Written Report, 10 pages (reference and non-literature review appendix are not counted as part of this page limit), in alignment with the Engineering Technical Report Style Guide.

Overview

You will develop a Project Proposal: Scope of Work for your Research Project incorporating a Literature Review and an overall plan for how you will work towards completion of your project. During this task, you will be expected to demonstrate professional conduct as a Student Engineer and in alignment with the Engineers Australia, Stage 1 Competency Standard for Professional Engineer and the Engineers Australia Code of Ethics.

What you will do

You have been given broad guidance as to the expected outcomes of your Research Project. To manage the project to completion, you are tasked with creating a Project Proposal: Scope of Work, at a similar standard to a Professional Engineer. To complete the Project Proposal: Scope of Work you will:

- Refine the overall Project Objective in consultation with your supervisor and any clients, industry partners or stakeholders. If you will need resources such as consumables or access to workshops and equipment, you should specify this.
- Conduct a review of the current state of engineering within the discipline or focus of your project, by reviewing research literature and current practice. You should attach a literature review to your Project Proposal: Scope of Work (no more than 5 pages words).
- Determine the deliverables that you will produce by the end of your project. This may include constructed engineering systems or models, reports, drawings, designs, reviews, or presentations. Note that you should include interim deliverables including status reports to your supervisor, clients, industry partners or stakeholders as required or testing rigs or models.
- Identify any risks, requirements, and constraints, including any relevant legislation, codes or regulations that will impact your engineering work.
- Agree with your supervisor and any clients, industry partners or stakeholders the modes of testing or validating your project outcomes to ensure appropriate quality standards are achieved. Note that it is important to build in interim quality check points, not just rely on testing or validating your final outcome.
- Determine the ways in which engineering for sustainability will be achieved in the conduct and outcome of your project. You may wish to refer to Sustainable Development Goals or Sustainability Standards which apply to your project (including Life Cycle analysis).
- Taking account of the above considerations, you will create a Project Timeline incorporating deliverables, dependencies, and milestones. You may wish to use the Project Proposal: Scope of Work template. Your submission must adhere to the Engineering Technical Report Style Guide.

For this assessment and the entirety of your project you are expected to engage in professional conduct in alignment with the Engineers Australia Code of Ethics.

What you will submit/present (what artefacts will be used to judge your performance?)

You can either submit to the *Project Proposal: Scope of Work Template* provided or use an alternative means of documenting your work, according to documentation guidelines elsewhere specified (such as a workplace or industry standard documentation). Regardless of which documentation approach you take, you **MUST** ensure that your report formatting meets the Engineering Technical Report Style Guide and addressed all the areas identified for analysis in relation to engineering practice. Your submission may not be marked by your supervisor, so you should ensure that it is written for a general engineering audience.

Criteria for Assessment

There are four criterion that will be used to assess your submission, all are equally weighted:

- Literature Review quality

- Analysis of project (risk, requirements, quality, sustainability, ethics)
- Appropriateness of proposed project approach (including methodology and planning)
- Professional engineering conduct (including professional communication)

The attached marking rubric outlines the standards required to achieve a grade against each criterion. Note that in class lectures and meetings with your supervisor will provide you with details of how your mark will be determined.

Resources and Support Available

Support

Library: Attend week 3 lectures and then contact your Liaison Librarian

Student Success: Book an appointment or attend a workshop at qut.to/academicsupport

Resources

Engineering Technical Report Style Guide

Library Guide: <https://libguides.library.qut.edu.au/engcap>

Writing a literature review in Engineering: accessible on your Unit Canvas site

(https://rise.articulate.com/share/v2lj-cMeDv5y4Pitc7NGpdDMXbgjiqz_)

Academic Integrity Statement

You are reminded that maintaining academic integrity is an important professional capability in all assessment. See the QUT [MOPP C/5.3 Academic Integrity](#) and the Engineers Australia [Code of Ethics](#) and Guidelines on Professional Conduct. For this assessment academic integrity will be monitored via:

- Interactions with your supervisor where you should discuss the details of your project and the references you are investigating.
- Plagiarism detection during submission via Turnitin.
- Verifying references through the marking process.

In finalising your mark, you may be asked to verify aspects of your submission through an interactive oral. Note that if the Unit Coordinator has reasonable concerns about your academic integrity, they may ask you to authenticate your learning in line with MOPP C/5.3 Section 5.3.7 (e) *Authentication of learning*.